

The empirical recurrence for OEIS A282313: recovering a conjecture that is not written down

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2 September 2026

Abstract

OEIS A282313 records “Empirical recurrence of order 57”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 87$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 57, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 57 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A282313 is “Number of nX5 0..1 arrays with no 1 equal to more than four of its king-move neighbors.”. It has offset 1 and begins

32, 984, 21055, 535494, 13473987, 335154112, 8386104536, 209686247202, ...

The entry states:

Empirical recurrence of order 57 (see link above)

As of the “Last modified” line on the live entry (Feb 11 2017 11:26:34, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1016$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 87$ of the 1016, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 87$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 174$ exact terms of the model returns order 57 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{57} c_i a(n-i),$$

c_1	16	c_{16}	40616869106	c_{31}	−13394907312869	c_{46}	−1948824767
c_2	152	c_{17}	62546217118	c_{32}	−4368851548859	c_{47}	1992794142
c_3	2026	c_{18}	−166639573219	c_{33}	2797246205461	c_{48}	1687258456
c_4	−3113	c_{19}	−383539341453	c_{34}	8400823720906	c_{49}	425782434
c_5	−31922	c_{20}	−571705048610	c_{35}	1184170479455	c_{50}	86501332
c_6	−412613	c_{21}	676636001111	c_{36}	−1495379879793	c_{51}	−49295866
c_7	−151540	c_{22}	1936003801750	c_{37}	−2771334064492	c_{52}	−18065004
c_8	1611860	c_{23}	2915692177202	c_{38}	−313092466852	c_{53}	−6847692
c_9	30395586	c_{24}	−766379800791	c_{39}	122401061601	c_{54}	695320
c_{10}	40603655	c_{25}	−5546151022301	c_{40}	538596834693	c_{55}	434640
c_{11}	16237709	c_{26}	−7686202366255	c_{41}	137428599235	c_{56}	164992
c_{12}	−1035518266	c_{27}	−1476825660811	c_{42}	94872349157	c_{57}	7936
c_{13}	−1964954778	c_{28}	9595614198649	c_{43}	−46913522056		
c_{14}	−2778773864	c_{29}	7796331622690	c_{44}	−31397940162		
c_{15}	17994046161	c_{30}	101027195215	c_{45}	−25324810106		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 57, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $57 + 4$ terms removed returns order 57 as well, so $\deg q_{\min} = 57 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $57 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 57 that this sequence satisfies — and that family turns out to have exactly one member.

References

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